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MATHS MASTER _____

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CANDIDATE NUMBER

2023 Trial Examination

Form VI Mathematics Advanced

Tuesday 8th August 2023

8:40am

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.
- Remove the central staple: you should have this cover booklet with Section I, 4 booklets for Section II, and the multiple-choice answer sheet.

Total Marks: 100

Section I (10 marks) Questions 1 – 10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11 – 37

- Relevant mathematical reasoning and calculations are required.
- Answer the questions in this paper in the spaces provided.
- This section is divided into four parts. Extra writing paper is provided at the end of each part.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on the start of the separate section and on the multiple choice sheet.
- Place everything inside this question booklet.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- Candidature: 91 pupils

Writer: OMD

	Marks
Multiple Choice	
Part A	
Part B	
Part C	
Part D	
TOTAL	

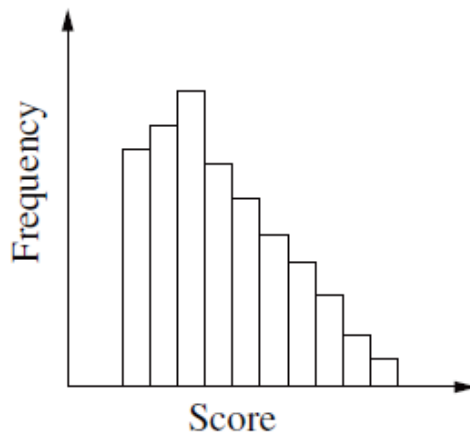
Section I

Questions in this section are multiple-choice.

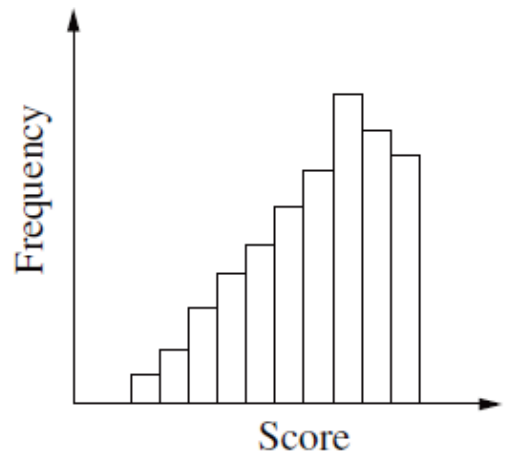
Record the single best answer for each question on the provided answer sheet.

1. Which histogram best represents a dataset that is positively skewed?

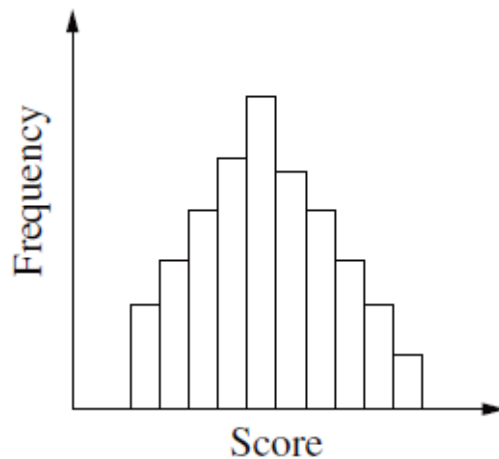
(A)



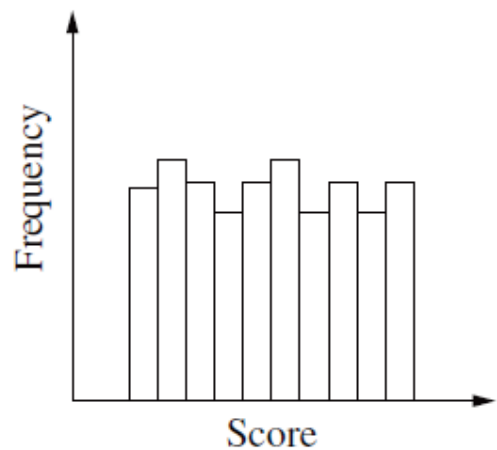
(B)



(C)



(D)



2. What is the gradient of the line passing through $A(3, 0)$ and $B(0, 4)$?

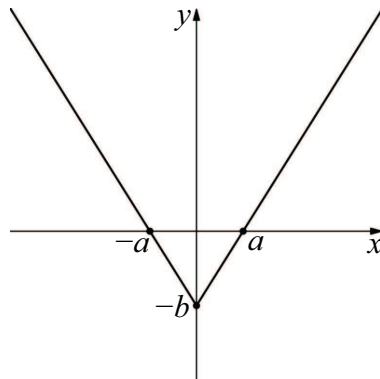
(A) $-\frac{4}{3}$

(B) $-\frac{3}{4}$

(C) $\frac{3}{4}$

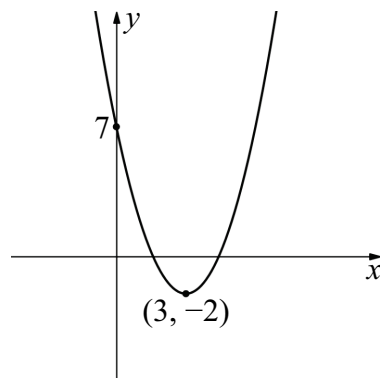
(D) $\frac{4}{3}$

3. Which type of symmetry best describes the function sketched below?



- (A) Odd
- (B) Even
- (C) Neither
- (D) Further information required

4. Which equation best describes the quadratic function sketched below?



- (A) $y = (x + 3)^2 + 2$
- (B) $y = (x + 3)^2 - 2$
- (C) $y = (x - 3)^2 + 2$
- (D) $y = (x - 3)^2 - 2$

5. In twelve years, the future value of an investment will be \$250 000. The interest rate is 6% per annum, compounded half-yearly.

Which equation will give the present value (PV) of the investment?

(A) $PV = \frac{250\,000}{(1 + 0.06)^{24}}$

(B) $PV = \frac{250\,000}{(1 + 0.03)^{24}}$

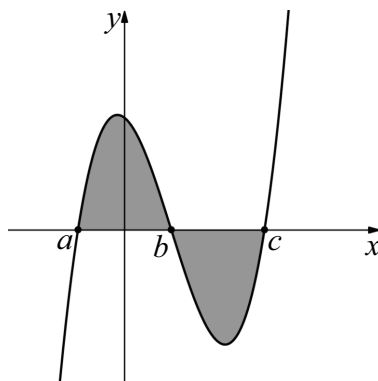
(C) $PV = \frac{250\,000}{(1 + 0.06)^{12}}$

(D) $PV = \frac{250\,000}{(1 + 0.03)^{12}}$

6. Which transformations best describe the change from $y = f(x)$ to $y = -f(x + 5)$?

- (A) A translation left 5 units then a reflection in the y -axis.
(B) A translation right 5 units then a reflection in the y -axis.
(C) A translation left 5 units then a reflection in the x -axis.
(D) A translation right 5 units then a reflection in the x -axis.

7. For the given function $y = f(x)$ sketched below, which integral does **NOT** give the shaded area?



- (A) $\int_a^b f(x) dx + \int_c^b f(x) dx$
- (B) $\int_a^b f(x) dx + \int_b^c f(x) dx$
- (C) $\int_a^b f(x) dx - \int_b^c f(x) dx$
- (D) $\int_a^b f(x) dx + \left| \int_b^c f(x) dx \right|$
8. For the discrete random variable X , it is known that $E(X) = 3$ and $\text{Var}(X) = 2$. Which option below gives the correct expected value and variance of $X + 3$?
- (A) $E(X + 3) = 3$ and $\text{Var}(X + 3) = 2$.
- (B) $E(X + 3) = 6$ and $\text{Var}(X + 3) = 2$.
- (C) $E(X + 3) = 3$ and $\text{Var}(X + 3) = 5$.
- (D) $E(X + 3) = 6$ and $\text{Var}(X + 3) = 5$.

9. Which is a correct trigonometric identity?

(A) $(1 + \sec \theta)(1 - \sec \theta) = \cot^2 \theta$

(B) $(1 + \sec \theta)(1 - \sec \theta) = \tan^2 \theta$

(C) $(1 + \sec \theta)(1 - \sec \theta) = -\cot^2 \theta$

(D) $(1 + \sec \theta)(1 - \sec \theta) = -\tan^2 \theta$

10. A region in the plane is bounded by the curve $y = \frac{1}{3x}$, the x -axis, the line $x = m$, and the line $x = 2m$, where $m > 0$. Which description most accurately describes the area of this region?

(A) The area increases as m increases.

(B) The area decreases as m increases.

(C) The area increases as m decreases.

(D) The area is independent of m .

End of Section I

The paper continues in the next section

SYDNEY GRAMMAR SCHOOL



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- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

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CANDIDATE NUMBER

Section II

Part A

— Remember to fill in your candidate number above —

QUESTION ELEVEN (2 marks)

Marks

Find the equation of the line perpendicular to $y = 2x - 1$ passing through the point $A(0, -7)$.

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QUESTION TWELVE (2 marks)

Marks

Classify each relation as one-to-one, many-to-one, one-to-many, or many-to-many.

(a) $y = 3x - 4$

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(b) $x^2 + y^2 = 25$

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QUESTION THIRTEEN (1 mark)

Marks

Differentiate $f(x) = x^2(2x - 1)$.

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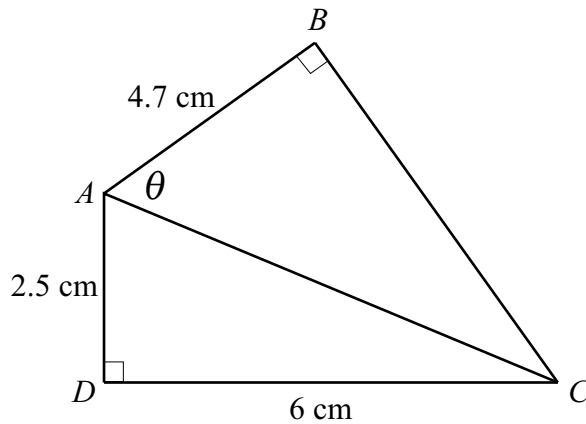
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QUESTION FOURTEEN (3 marks)

Marks

3Two right-angled triangles, ABC and ADC , are shown below.Calculate the size of angle θ . Give your answer correct to the nearest minute.

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QUESTION FIFTEEN (2 marks)

Marks

Suppose that $f(x) = 2x^2$ and $g(x) = x - 1$.(a) Find the value of $g(f(4))$.**1**

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(b) Find $f(g(x))$.**1**

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QUESTION SIXTEEN (2 marks)

Marks

Find a formula for the n th term of the following sequences:

(a) 10, 12, 14, 16, ...

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(b) $\frac{1}{3}, 1, 3, 9, \dots$

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QUESTION SEVENTEEN (2 marks)

Marks

Differentiate $y = (5x - 3)^2$.

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QUESTION EIGHTEEN (2 marks)

Marks

The gradient function of a curve is given by $f'(x) = 2x$. Given that the curve $y = f(x)$ goes through the point $A(2, -1)$, find the equation of the curve, $y = f(x)$.

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QUESTION NINETEEN (4 marks)**Marks**

Daniel inherits \$50 000 and invests it in an account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Daniel withdraws \$700.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.006) - 700,$$

where $n = 1, 2, 3, \dots$ and $A_0 = 50\,000$.

- (a) Use the recurrence relation to find the amount of money in the account immediately after the third withdrawal. 2

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- (b) Calculate the amount of interest earned in the first three months. 2

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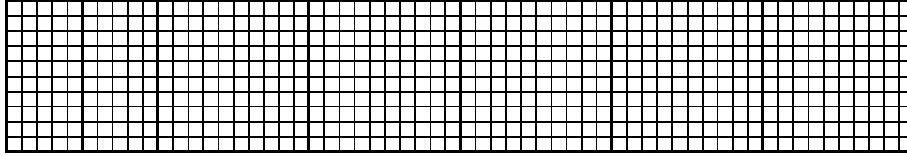
QUESTION TWENTY (4 marks)**Marks**

The Jawas run a trading outpost on Tatooine. Their accountant records their monthly sales, in millions of galactic credits, listed below:

3, 4, 7, 9, 11, 12, 13, 13, 14, 20, 25.

- (a) On the graph paper below, draw a box plot to represent this data.

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- (b) Identify any outliers in this data.

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- (c) The Jawas claim that for 75% of the months, the amount received in sales is greater than 10 million galactic credits. Comment on this claim, justifying your answer.

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CANDIDATE NUMBER

Section II

Part B

— Remember to fill in your candidate number above —

QUESTION TWENTY-ONE (3 marks)

Marks

(a) Find the equation of the tangent to the curve $y = e^{x+2}$ at the point $A(-2, 1)$.

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(b) Hence find point B , where the tangent in Part (a) crosses the x -axis.

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QUESTION TWENTY-TWO (3 marks)

Marks

Solve the equation $9^x - 3^{x+1} = -2$, leave your answers in exact form.

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QUESTION TWENTY-THREE (2 marks)

Marks

Find the sum of the first 100 positive odd numbers.

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QUESTION TWENTY-FOUR (2 marks)

Marks

Differentiate $y = \frac{x^2 + 1}{e^{2x}}$, leave your answer in simplest form.

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QUESTION TWENTY-FIVE (4 marks)

Marks

The table shows the future value of an annuity of \$1.

Future values of an annuity of \$1				
Years	Interest Rate per Annum			
	1%	2%	3%	4%
4	4.060	4.122	4.184	4.246
5	5.101	5.204	5.309	5.416
6	6.152	6.308	6.468	6.633

Harper is saving for a trip and estimates she will need \$20 000. She opens an account earning 4% per annum, compounded annually.

- (a) How much does Harper need to deposit every year if she wishes to have enough money for the trip in 5 years time? 2

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- (b) How much interest will Harper earn on her investment over the 5 years? Give your answer correct to the nearest dollar. 2

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QUESTION TWENTY-SIX (3 marks)**Marks**

(a) Find $\int e^{3-2x} dx$.

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(b) Evaluate $\int_0^1 \frac{1}{3x+1} dx$, leave your answer in exact form.

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QUESTION TWENTY-SEVEN (3 marks)

Marks

Gizka have proven to be a pest throughout the galaxy due to their exponential rate of reproduction. The number, G , of gizka at a time, t , months after they were introduced to the planet of Tatooine can be calculated by the formula $G = 50e^{1.2t}$.

(a) How many gizka were first released on Tatooine? 1

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(b) Show that $\frac{dG}{dt} = 1.2G$ and hence find the rate at which the number of gizka was increasing when there were 200 gizka. 2

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QUESTION TWENTY-EIGHT (2 marks)

Marks

(a) Differentiate $\cos^3 x$ with respect to x .

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(b) Hence, or otherwise, find $\int \sin x \cos^2 x \, dx$.

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CANDIDATE NUMBER

Section II

Part C

— Remember to fill in your candidate number above —

QUESTION TWENTY-NINE (5 marks)

Marks

A bowl of fruit contains 18 apples of which 11 are red and 7 are green.

Matthew takes one apple at random and eats it. Evelyn then takes an apple at random and eats it.

- (a) By drawing a probability tree diagram, or otherwise, find the probability that Matthew and Evelyn eat apples of the same colour. 3

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- (b) Given that Matthew and Evelyn eat apples of the same colour, find the probability that they are red. 2

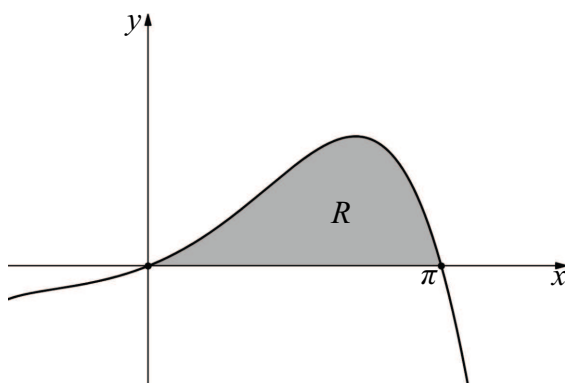
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QUESTION THIRTY (3 marks)**Marks**

The curve shown in the diagram above has the equation $y = \frac{e^x \sin x}{x + 2}$.

The finite region R bounded by the curve and the x -axis from $x = 0$ to $x = \pi$ is shown shaded in the diagram above.

- (a) Complete the table below with the value of y corresponding to $x = \frac{\pi}{3}$, giving your answer correct to 5 decimal places. 1

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π
y	0		1.71761	0

- (b) Use the trapezoidal rule, with all the values in the completed table, to obtain an estimate for $\int_0^\pi \frac{e^x \sin x}{x + 2} dx$, the area of the region R . Give your answer correct to 4 decimal places. 2

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QUESTION THIRTY-ONE (4 marks)

Marks

The partial sums S_n of a geometric progression are given by 100, 190, 271, . . .

- (a) By finding a formula for the n th term, T_n , find the value of n which gives the first term in the geometric progression smaller than 10. 3

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- (b) Find the limiting sum, S_∞ , of the geometric progression. 1

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QUESTION THIRTY-TWO (5 marks)

Marks

A function is defined as $f(x) = \sin\left(2x + \frac{\pi}{2}\right) + 1$, $0 \leq x \leq 2\pi$.

(a) Sketch $y = f(x)$, clearly labelling any stationary points and intercepts with the axes. 3

(b) Hence, or otherwise, solve $\sin\left(2x + \frac{\pi}{2}\right) = 1$, $0 \leq x \leq 2\pi$. 2

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QUESTION THIRTY-THREE (9 marks)

Marks

Let $f(x) = \frac{x^2 - 1}{x^3}$.

(a) State the natural domain of the function.

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(b) Show that the function is odd.

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It is given that $f'(x) = \frac{-x^2 + 3}{x^4}$.

(c) Show that $f''(x) = \frac{2x^2 - 12}{x^5}$.

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QUESTION THIRTY-THREE (Continued)

(d) Find any stationary points of $f(x)$ and determine their nature. 2

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(e) Sketch the curve $y = \frac{x^2 - 1}{x^3}$, clearly labelling any stationary points, asymptotes, and intercepts with the axes. 4

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CANDIDATE NUMBER

Section II

Part D

— Remember to fill in your candidate number above —

QUESTION THIRTY-FOUR (3 marks)

Marks

Solve $2\log_2(x + 3) - \log_2 x = 4$.

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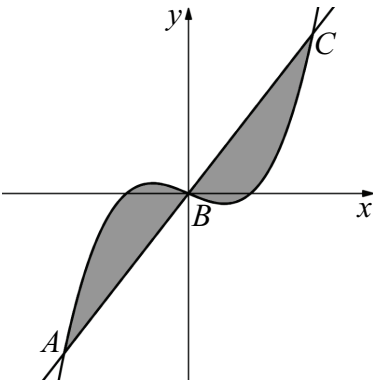
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QUESTION THIRTY-FIVE (4 marks)

Marks

4



Given $f(x) = x^3 - x$ and $g(x) = 3x$, find the area enclosed by $y = f(x)$ and $y = g(x)$, as shown by the shaded regions in the diagram above.

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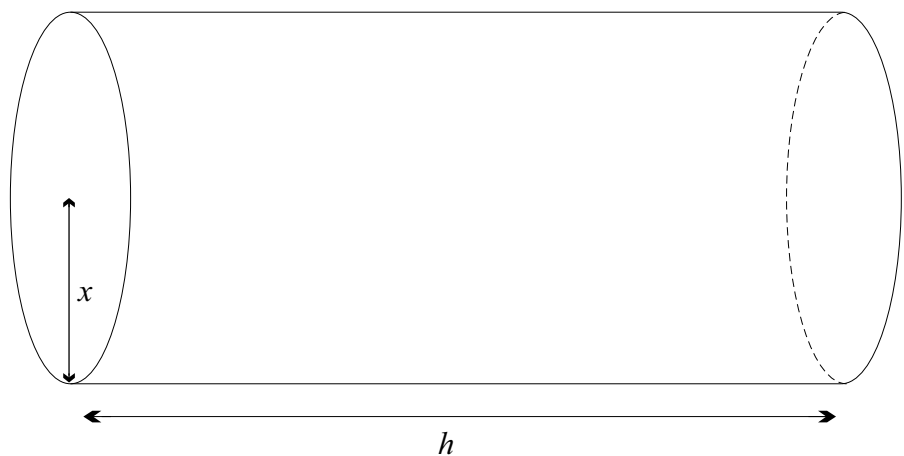
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QUESTION THIRTY-SIX (6 marks)

Marks



The diatium power cell is an integral component in all lightsabers. Each cell is cylindrical in shape, with a base radius x cm and height h cm, as shown in the diagram above.

- (a) Given that the volume of each cell has to be 60 cm^3 , show that the surface area, $A\text{ cm}^2$, of the cell is given by $A = 2\pi x^2 + \frac{120}{x}$. 2

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QUESTION THIRTY-SIX (Continued)

- (b) For safety reasons it is imperative that the surface area is as low as possible. Calculate the minimum surface area, correct to 2 decimal places, and justify your answer.

4

[illegible]

QUESTION THIRTY-SEVEN (5 marks)

Marks

Bo-Katan and Din are racing with their jetpacks along a 2 km linear course. Bo-Katan accelerates at a constant 10 m/s^2 up to a top speed of 35 m/s , which she maintains until the finish. Din accelerates at a constant 8 m/s^2 up to a top speed of 40 m/s , which he maintains until the finish.

- (a) Due to a technical malfunction, Din starts n seconds after Bo-Katan. Given that they both start from rest, and Din wins the race, what is the largest possible value of n ? Give your answer correct to 2 decimal places.

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SOLUTIONS

NAME _____

MATHS MASTER _____

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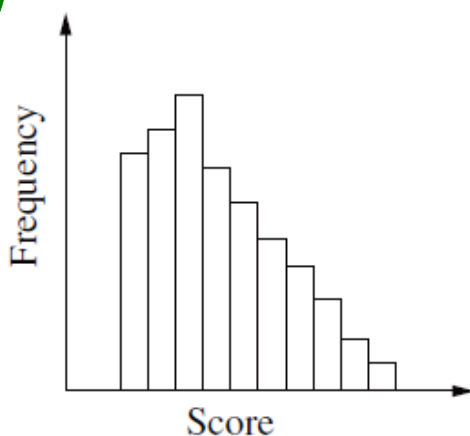
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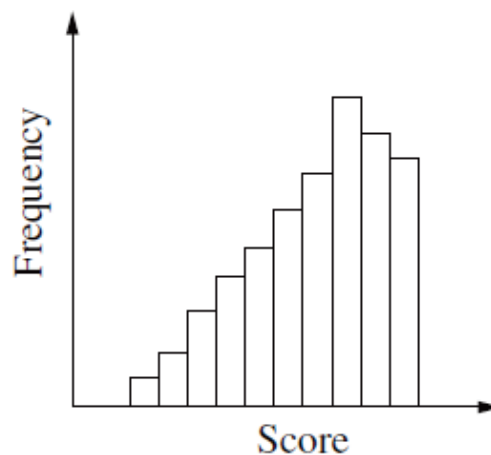
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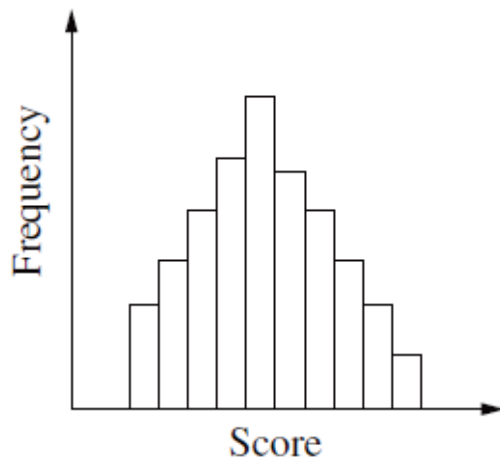
(A)



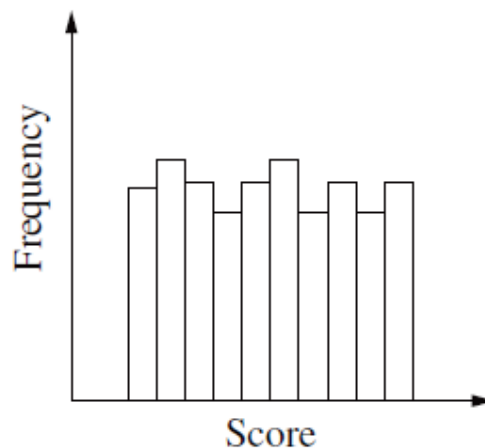
(B)



(C)



(D)



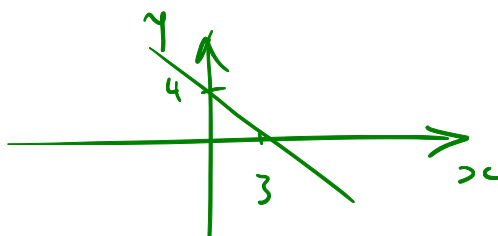
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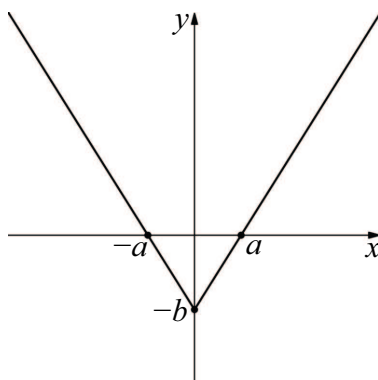
(C) $\frac{3}{4}$

(D) $\frac{4}{3}$



$$m = -\frac{4}{3}$$

3. Which type of symmetry best describes the function sketched below?



(A) Odd

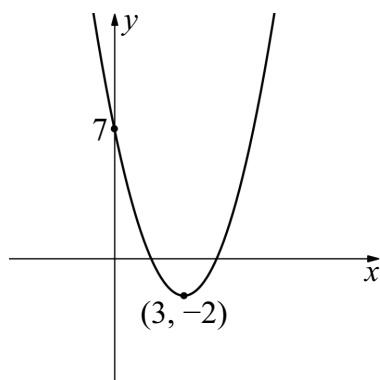
(B) Even

(C) Neither

(D) Further information required

line symmetry $x=0$.

4. Which equation best describes the quadratic function sketched below?



(A) $y = (x + 3)^2 + 2$

(B) $y = (x + 3)^2 - 2$

(C) $y = (x - 3)^2 + 2$

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right 3 down 2

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(B) $PV = \frac{250\,000}{(1 + 0.03)^{24}}$

(C) $PV = \frac{250\,000}{(1 + 0.06)^{12}}$

(D) $PV = \frac{250\,000}{(1 + 0.03)^{12}}$

half-yearly
 $12 \text{ yrs} \times 2 = 24 \text{ half years}$
6% per annum
 $\frac{6\%}{2} = 3\% \text{ per half year}$

6. Which transformations best describe the change from $y = f(x)$ to $y = -f(x + 5)$?

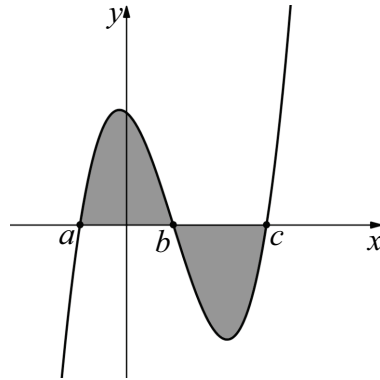
(A) A translation left 5 units then a reflection in the y -axis.

(B) A translation right 5 units then a reflection in the y -axis.

(C) A translation left 5 units then a reflection in the x -axis.

(D) A translation right 5 units then a reflection in the x -axis.

7. For the given function $y = f(x)$ sketched below which integral does **NOT** give the shaded area?



- (A) $\int_a^b f(x) dx + \int_c^b f(x) dx$
- (B) $\int_a^b f(x) dx + \int_b^c f(x) dx$
- (C) $\int_a^b f(x) dx - \int_b^c f(x) dx$
- (D) $\int_a^b f(x) dx + \left| \int_b^c f(x) dx \right|$

8. For the random variable X , it is known that $E(X) = 3$ and $\text{Var}(X) = 2$. Which option below gives the correct expected value and variance of $X + 3$?

- (A) $E(X + 3) = 3$ and $\text{Var}(X + 3) = 2$.
- (B) $E(X + 3) = 6$ and $\text{Var}(X + 3) = 2$.
- (C) $E(X + 3) = 3$ and $\text{Var}(X + 3) = 5$.
- (D) $E(X + 3) = 6$ and $\text{Var}(X + 3) = 5$.

$$E(aX + b) = aE(X) + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

9. Which is a correct trigonometric identity?

(A) $(1 + \sec \theta)(1 - \sec \theta) = \cot^2 \theta$

(B) $(1 + \sec \theta)(1 - \sec \theta) = \tan^2 \theta$

(C) $(1 + \sec \theta)(1 - \sec \theta) = -\cot^2 \theta$

(D) $(1 + \sec \theta)(1 - \sec \theta) = -\tan^2 \theta$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 - \sec^2 \theta = -\tan^2 \theta$$

$$(1 + \sec \theta)(1 - \sec \theta) = -\tan^2 \theta$$

10. A region in the plane is bounded by the curve $y = \frac{1}{3x}$, the x -axis, the line $x = m$, and the line $x = 2m$, where $m > 0$. Which description most accurately describes the area of this region?

(A) The area increases as m increases.

(B) The area decreases as m increases.

(C) The area increases as m decreases.

(D) The area is independent of m .

$$\frac{1}{3} \int_m^{2m} \frac{1}{x} dx$$

$$= \frac{1}{3} [\ln x]_m^{2m}$$

$$= \frac{1}{3} (\ln 2m - \ln m)$$

$$= \frac{1}{3} \ln \frac{2m}{m}$$

$$= \frac{1}{3} \ln 2$$

End of Section I

The paper continues in the next section

QUESTION ELEVEN (2 marks)

Marks

Find the equation of the line perpendicular to $y = 2x - 1$ passing through the point $A(0, -7)$.

2

$$\begin{aligned}
 m_1 &= 2 \\
 m_2 &= -\frac{1}{2} \quad \checkmark \\
 y - (-7) &= -\frac{1}{2}(x - 0) \\
 y + 7 &= -\frac{1}{2}x \\
 y &= -\frac{1}{2}x - 7 \quad \checkmark
 \end{aligned}$$

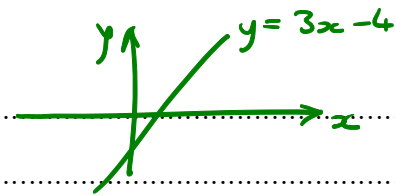
QUESTION TWELVE (2 marks)

Marks

Classify each relation as one-to-one, many-to-one, one-to-many, or many-to-many.

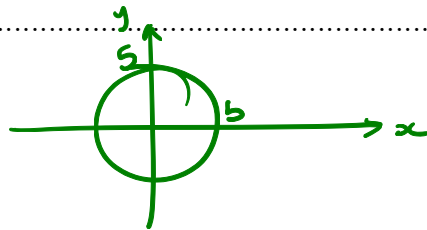
(a) $y = 3x - 4$

1

one-to-one $\checkmark$ 

(b) $x^2 + y^2 = 25$

1

many-to-many $\checkmark$ **QUESTION THIRTEEN** (1 mark)

Marks

Differentiate $f(x) = x^2(2x - 1)$.

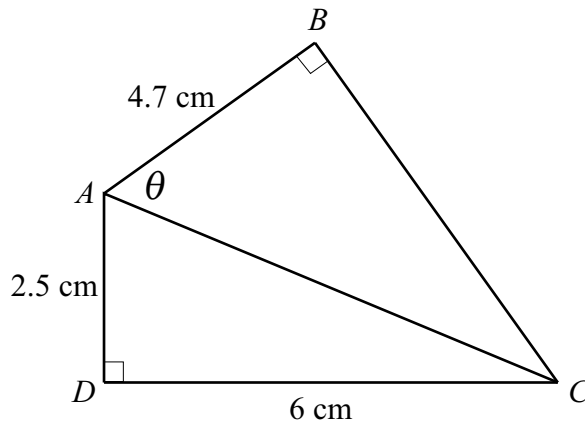
1

$$f(x) = 2x^3 - x^2$$

$$f'(x) = 6x^2 - 2x \quad \checkmark$$

QUESTION FOURTEEN (3 marks)

Marks

3Two right-angled triangles, ABC and ADC , are shown below.Calculate the size of angle θ . Give your answer correct to the nearest minute.

$$AC = \sqrt{6^2 + 2.5^2}$$

$$= 6.5 \text{ cm}$$

$$\theta = \cos^{-1} \left(\frac{4.7}{6.5} \right)$$

$$= 43^\circ 41'$$

QUESTION FIFTEEN (2 marks)

Marks

1Suppose that $f(x) = 2x^2$ and $g(x) = x - 1$.(a) Find the value of $g(f(4))$.

$$f(4) = 32$$

$$g(32) = 31$$

$$\therefore g(f(4)) = 31$$

(b) Find $f(g(x))$.**1**

$$f(g(x)) = 2(x-1)^2$$

$$= 2(x^2 - 2x + 1)$$

$$= 2x^2 - 4x + 2$$

accept factorised form

QUESTION SIXTEEN (2 marks)

Marks

Find a formula for the n th term of the following sequences:

(a) 10, 12, 14, 16, ...

1

$$\begin{aligned} \text{A.P. } T_n &= a + (n-1)d \quad a = 10 \quad d = 2 \\ &= 10 + 2(n-1) \quad \checkmark \text{ accept} \\ &= 2n + 8 \end{aligned}$$

(b) $\frac{1}{3}, 1, 3, 9, \dots$

1

$$\begin{aligned} \text{G.P. } T_n &= ar^{n-1} \quad a = \frac{1}{3} \quad r = 3 \\ &= \frac{1}{3} \times 3^{n-1} \quad \checkmark \text{ accept} \\ &= 3^{n-2} \end{aligned}$$

QUESTION SEVENTEEN (2 marks)

Marks

Differentiate $y = (5x - 3)^2$.

2

$$y = (5x - 3)^2$$

$$y' = 2 \times 5 \times (5x - 3) \checkmark$$

$$y' = 10(5x - 3) \checkmark$$

QUESTION EIGHTEEN (2 marks)

Marks

The gradient function of a curve is given by $f'(x) = 2x$. Given that the curve $y = f(x)$ goes through the point $A(2, -1)$, find the equation of the curve, $y = f(x)$.

2

$$f'(x) = 2x$$

$$f(x) = x^2 + c \checkmark$$

$$-1 = 2^2 + c$$

$$-1 = 4 + c$$

$$c = -5$$

$$f(x) = x^2 - 5 \checkmark$$

QUESTION NINETEEN (4 marks)

Marks

Daniel inherits \$50 000 and invests it in an account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Daniel withdraws \$700.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.006) - 700,$$

where $n = 1, 2, 3, \dots$ and $A_0 = 50\,000$.

- (a) Use the recurrence relation to find the amount of money in the account immediately after the third withdrawal. 2

$$\begin{aligned} A_1 &= 50000(1.006) - 700 \\ &= 49600 \end{aligned}$$

$$\begin{aligned} A_2 &= 49600(1.006) - 700 \\ &= 49197.6 \end{aligned}$$

$$\begin{aligned} A_3 &= 49197.6(1.006) - 700 \\ &= \$48792.79 \end{aligned}$$

- (b) Calculate the amount of interest earned in the first three months. 2

$$\begin{aligned} I &= A_3 + 3 \times 700 - 50000 \\ &= \$892.79 \end{aligned}$$

Allow ECF from Part (a)
their $A_3 + 3 \times 700 - 50000$

QUESTION TWENTY (4 marks)

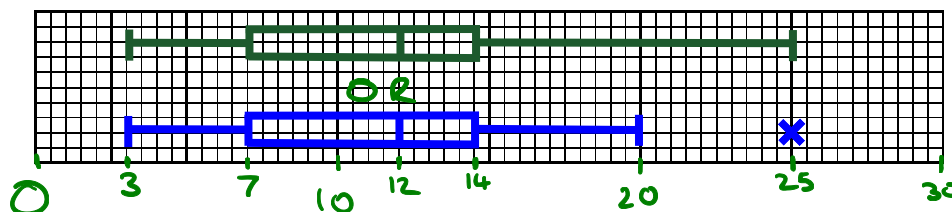
Marks

The Jawas run a trading outpost on Tatooine. Their accountant records their monthly sales, in millions of galactic credits, listed below:

3, 4, 7, 9, 11, 12, 13, 13, 14, 20, 25.

- (a) On the graph paper below, draw a box plot to represent this data.

2



$Q_1 + Q_2 + Q_3$ ✓
whiskers ✓

3, 4, 7, 9, 11, 12, 13, 13, 14, 20, 25
 Q_1 Q_2 Q_3

- (b) Identify any outliers in this data.

1

$$IQR = 14 - 7 = 7$$

$$\text{Lower} = 7 - 7 \times 1.5 = -3.5$$

$\therefore$ no outliers at bottom end

$$\text{Upper} = 14 + 7 \times 1.5 = 24.5$$

$\therefore$ 25 outlier ✓

- (c) The Jawas claim that for 75% of the months, the amount received in sales is greater than 10 million galactic credits. Comment on this claim, justifying your answer.

1

This claim is false. 75% of the months are above the lower quartile (Q_1) which is 7 million galactic credits.

reference to Q_1
or 7 million galactic credits ✓

QUESTION TWENTY-ONE (3 marks)

Marks

- (a) Find the equation of the tangent to the curve
- $y = e^{x+2}$
- at the point
- $A(-2, 1)$
- .

2

$$y' = e^{x+2}$$

$$\text{when } x = -2$$

$$y' = e^{-2+2}$$

$$= e^0$$

$$y' = 1$$

$$y - 1 = 1(x - -2)$$

$$y - 1 = x + 2$$

$$y = x + 3$$

- (b) Hence find point
- B
- , where the tangent in Part (a) crosses the
- x
- axis.

1

$$\text{when } y = 0$$

$$0 = x + 3$$

$$x = -3$$

$$(-3, 0)$$

✓ accept either

QUESTION TWENTY-TWO (3 marks)

Marks

Solve the equation $9^x - 3^{x+1} = -2$, leave your answers in exact form.

3

$$(3^x)^2 - 3(3^x) + 2 = 0$$

$$\text{Let } y = 3^x$$

$$y^2 - 3y + 2 = 0$$

$$(y-2)(y-1) = 0$$

$$\therefore \text{ EITHER } y-2=0 \quad \text{OR} \quad y-1=0$$

$$y=2$$

$$3^x = 2$$

$$x = \log_3 2$$

$$\doteq 0.6309$$

$$y=1$$

$$3^x = 1$$

$$x = 0$$

QUESTION TWENTY-THREE (2 marks)

Marks

Find the sum of the first 100 positive odd numbers.

2

$$\text{A.P. } a=1 \quad d=2 \quad n=100$$

$$S_{100} = \frac{100}{2} [2 \times 1 + (100-1) \times 2] \quad \checkmark$$

$$= 10\,000 \quad \checkmark$$

QUESTION TWENTY-FOUR (2 marks)

Marks

Differentiate $y = \frac{x^2 + 1}{e^{2x}}$, leave your answer in simplest form.

2

$$y = \frac{u}{v}$$

$$u = x^2 + 1$$

$$v = e^{2x}$$

$$u' = 2x$$

$$v' = 2e^{2x}$$

$$y' = \frac{v u' - u v'}{v^2}$$

$$= \frac{2x e^{2x} - 2e^{2x}(x^2 + 1)}{e^{4x}} \quad \checkmark$$

$$= \frac{2x - 2(x^2 + 1)}{e^{2x}}$$

$$= \frac{2x - 2x^2 - 2}{e^{2x}} \quad \checkmark$$

QUESTION TWENTY-FIVE (4 marks)

Marks

The table shows the future value of an annuity of \$1.

Future values of an annuity of \$1				
Years	Interest Rate per Annum			
	1%	2%	3%	4%
4	4.060	4.122	4.184	4.246
5	5.101	5.204	5.309	5.416
6	6.152	6.308	6.468	6.633

Harper is saving for a trip and estimates she will need \$20 000. She opens an account earning 4% per annum, compounded annually.

- (a) How much does Harper need to deposit every year if she wishes to have enough money for the trip in 5 years time? 2

$$\begin{aligned}
 FV. &= 20000 \div (5.416) \\
 &= \$3692.76
 \end{aligned}$$

allow ECF

- (b) How much interest will Harper earn on her investment over the 5 years? Give your answer correct to the nearest dollar. 2

$$\begin{aligned}
 I &= 20000 - 5 \times 3692.76 \\
 &= \$1536
 \end{aligned}$$

allow ECF

QUESTION TWENTY-SIX (3 marks)

Marks

(a) Find $\int e^{3-2x} dx$.

1

$$-\frac{1}{2} \int -2 e^{3-2x} dx$$

$$-\frac{1}{2} e^{3-2x} (+C)$$

(b) Evaluate $\int_0^1 \frac{1}{3x+1} dx$, leave your answer in exact form.

2

$$\frac{1}{3} \int_0^1 \frac{3}{3x+1} dx$$

$$= \frac{1}{3} \left[\ln |3x+1| \right]_0^1$$

$$= \frac{1}{3} (\ln 4 - \ln 1)$$

$$= \frac{1}{3} \ln 4$$

QUESTION TWENTY-SEVEN (3 marks)

Marks

Gizka have proven to be a pest throughout the galaxy due to their exponential rate of reproduction. The number, G , of gizka at a time, t , months after they were introduced to the planet of Tatooine can be calculated by the formula $G = 50e^{1.2t}$.

- (a) How many gizka were first released on Tatooine?

1

when $t = 0$
 $G = 50$ ✓

- (b) Show that $\frac{dG}{dt} = 1.2G$ and hence find the rate at which the number of gizka was increasing when there were 200 gizka.

2

LHS = $\frac{dG}{dt}$
 $= 50e^{1.2t} \times 1.2$
 $= 60e^{1.2t}$
 $= \text{RHS}$

RHS = $1.2G$
 $= 1.2 \times 50e^{1.2t}$
 $= 60e^{1.2t}$

✓ show that

$\frac{dG}{dt} = 1.2 \times 200$

$= 240$ gizka per month ✓

QUESTION TWENTY-EIGHT (2 marks)

Marks

- (a) Differentiate
- $\cos^3 x$
- with respect to
- x
- .

1

$$\begin{aligned}y &= \cos^3 x \\ &= u^3 \\ \frac{dy}{du} &= 3u^2 \\ &= 3\cos^2 x \\ u &= \cos x \\ \frac{du}{dx} &= -\sin x \\ \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= -3\sin x \cos^2 x \quad \checkmark\end{aligned}$$

- (b) Hence, or otherwise, find
- $\int \sin x \cos^2 x \, dx$
- .

1

$$\begin{aligned}& -\frac{1}{3} \int -3\sin x \cos^2 x \, dx \\ &= -\frac{1}{3} \cos^3 x + C \quad \checkmark\end{aligned}$$

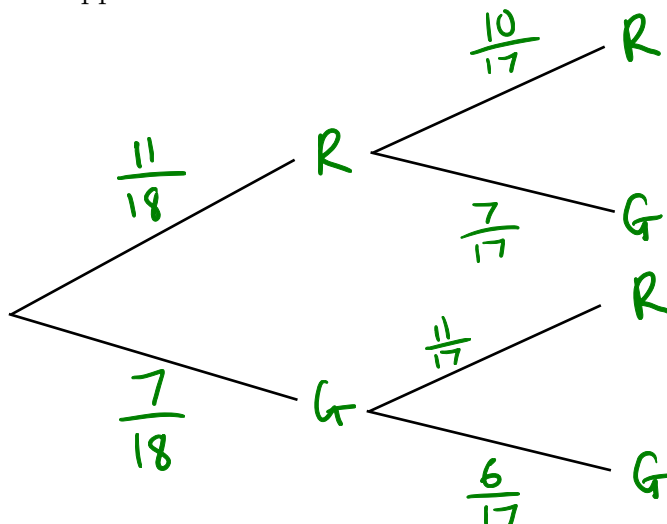
QUESTION TWENTY-NINE (5 marks)

Marks

A bowl of fruit contains 18 apples of which 11 are red and 7 are green.

Matthew takes one apple at random and eats it. Evelyn then takes an apple at random and eats it.

- (a) By drawing a probability tree diagram, or otherwise, find the probability that Matthew and Evelyn eat apples of the same colour. 3



$$P(\text{SAME}) = P(RR) + P(GG)$$

$$= \frac{11}{18} \times \frac{10}{17} + \frac{7}{18} \times \frac{6}{17}$$

 $P(RR)$ or
 $P(GG)$

$$= \frac{76}{153}$$

✓ allow ft from incorrect tree diagram.

- (b) Given that Matthew and Evelyn eat apples of the same colour, find the probability that they are red. 2

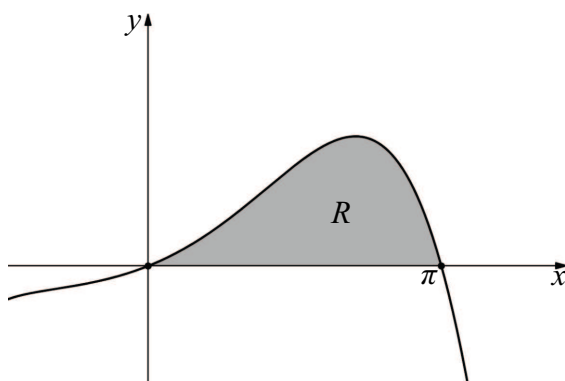
$$P(RR | \text{SAME}) = \frac{P(RR)}{P(\text{SAME})}$$

$$= \frac{55}{153} \div \frac{76}{153}$$

$$= \frac{55}{76}$$

QUESTION THIRTY (3 marks)

Marks



The curve shown in the diagram above has the equation $y = \frac{e^x \sin x}{x + 2}$.

The finite region R bounded by the curve and the x -axis from $x = 0$ to $x = \pi$ is shown shaded in the diagram above.

- (a) Complete the table below with the value of y corresponding to $x = \frac{\pi}{3}$, giving your answer correct to 5 decimal places. 1

DEG MODE = 0.01709122979

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π
y	0	0.80988	1.71761	0

- (b) Use the trapezoidal rule, with all the values in the completed table, to obtain an estimate for $\int_0^\pi \frac{e^x \sin x}{x + 2} dx$, the area of the region R . Give your answer correct to 4 decimal places. 2

$$R = \frac{\pi - 0}{2 \times 3} \left[0 + 0 + 2 \left(0.80988 + 1.71761 \right) \right]$$

✓ correct use of formula

$$= 2.646784199 \quad \text{with unrounded value of } f\left(\frac{\pi}{2}\right)$$

$$= 2.646781339 \quad \text{with rounded value of } f\left(\frac{\pi}{2}\right)$$

$$\div 2.6468 \text{ (4dp)} \quad \checkmark \text{ rounding}$$

DEG MODE value gives $\div 1.81657488$
✓ ECF

$\div 1.8166 \text{ (4dp)}$

QUESTION THIRTY-ONE (4 marks)

Marks

The partial sums S_n of a geometric progression are given by 100, 190, 271, ...

- (a) By finding a formula for the n th term, T_n , find the value of n which gives the first term in the geometric progression smaller than 10. 3

$$\begin{aligned}
 T_1 &= 100 \\
 T_2 &= 190 - 100 \\
 &= 90 \\
 T_3 &= 271 - 190 \\
 &= 81
 \end{aligned}
 \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \text{G.P. } a=100 \quad r=0.9$$

$$T_n = 100 \times 0.9^{n-1}$$

$$10 > 100 \times 0.9^{n-1}$$

$$0.1 > 0.9^{n-1}$$

$$\log(0.1) > (n-1) \log(0.9)$$

$$\frac{\log(0.1)}{\log(0.9)} < n-1$$

$$n > \frac{\log 0.1}{\log 0.9} + 1$$

$$n > 22.85$$

$$\therefore n = 23$$

- (b) Find the limiting sum, S_∞ , of the geometric progression. 1

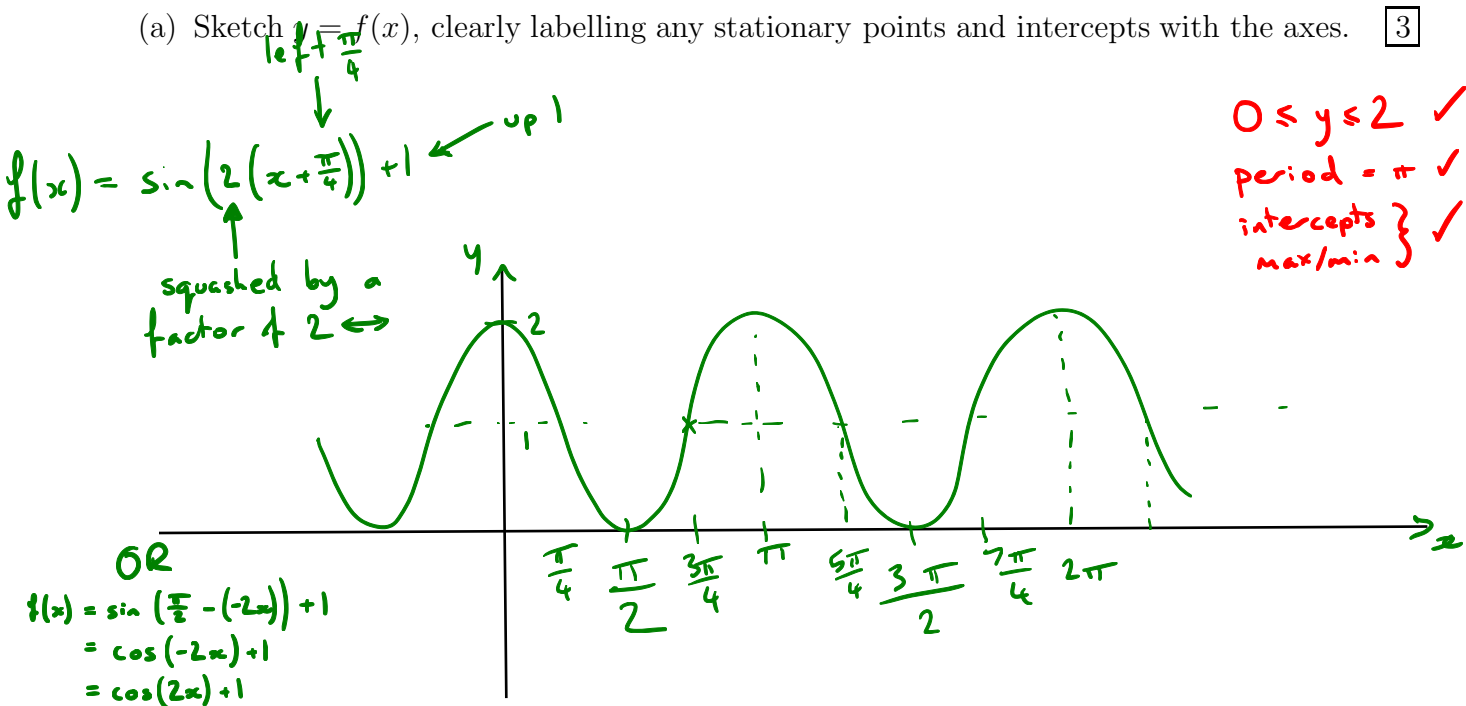
$$\begin{aligned}
 S_\infty &= \frac{100}{1-0.9} \\
 &= 1000
 \end{aligned}$$

QUESTION THIRTY-TWO (5 marks)

Marks

A function is defined as $f(x) = \sin\left(2x + \frac{\pi}{2}\right) + 1$, $0 \leq x \leq 2\pi$.

(a) Sketch $y = f(x)$, clearly labelling any stationary points and intercepts with the axes. 3



$0 \leq y \leq 2$ ✓
 period = π ✓
 intercepts } ✓
 max/min }

(b) Hence, or otherwise, solve $\sin\left(2x + \frac{\pi}{2}\right) = 1$, $0 \leq x \leq 2\pi$. 2

$$0 \leq 2x \leq 4\pi$$

$$\frac{\pi}{2} \leq 2x + \frac{\pi}{2} \leq \frac{9\pi}{2}$$

$$\sin\left(2x + \frac{\pi}{2}\right) = 1 \quad (+1)$$

$$\sin\left(2x + \frac{\pi}{2}\right) + 1 = 2$$

$$2x + \frac{\pi}{2} = \sin^{-1} 1$$

$$= \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} \quad \checkmark$$

$$\therefore \text{where graph } f(x) = 2 \quad \checkmark$$

$$2x = 0, 2\pi, 4\pi$$

$$x = 0, \pi, 2\pi \quad \checkmark$$

$$x = 0, \pi, 2\pi \quad \checkmark$$

allow ECF

graphical solution

or

algebraic solution

QUESTION THIRTY-THREE (9 marks)

Marks

Let $f(x) = \frac{x^2 - 1}{x^3}$.

- (a) State the natural domain of the function.

1

$$x \neq 0$$

- (b) Show that the function is odd.

1

$$\begin{aligned}
 -f(x) &= -\frac{x^2 - 1}{x^3} & f(-x) &= \frac{(-x)^2 - 1}{(-x)^3} \\
 & & &= \frac{x^2 - 1}{-x^3} \\
 -f(x) &= f(-x) \quad \therefore \text{ODD} & &= -\frac{x^2 - 1}{x^3}
 \end{aligned}$$

show that

It is given that $f'(x) = \frac{-x^2 + 3}{x^4}$.

- (c) Show that $f''(x) = \frac{2x^2 - 12}{x^5}$.

1

$$f'(x) = -x^{-2} + 3x^{-4}$$

$$f''(x) = 2x^{-3} - 12x^{-5}$$

$$= \frac{2x^2 - 12}{x^5} \quad \text{As required} \quad \text{show that}$$

QUESTION THIRTY-THREE (Continued)(d) Find any stationary points of $f(x)$ and determine their nature.

2

$$f'(x) = \frac{(\sqrt{3}+x)(\sqrt{3}-x)}{x^4} \quad \text{s.p. when } x = \sqrt{3} \text{ or } -\sqrt{3} \quad \checkmark$$

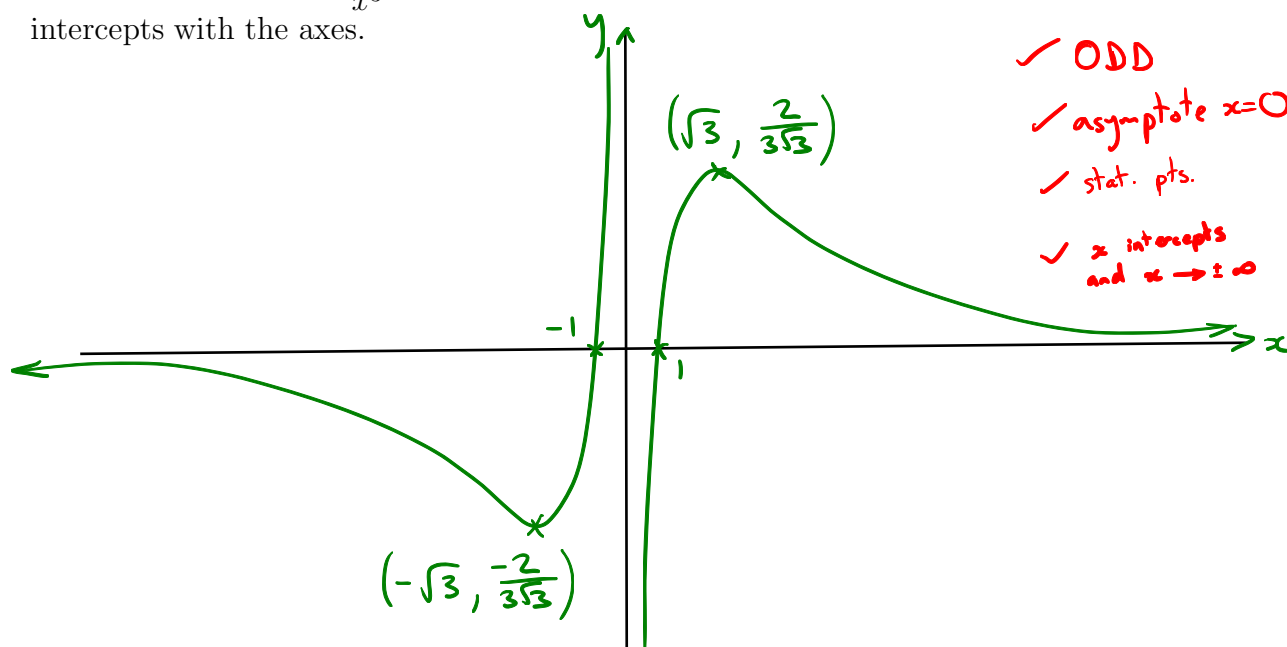
$$y = \frac{2}{3\sqrt{3}} \quad y = \frac{-2}{3\sqrt{3}}$$

$$f''(\sqrt{3}) = \frac{-6}{9\sqrt{3}} \quad f''(-\sqrt{3}) = \frac{6}{9\sqrt{3}}$$

$$\begin{array}{cc} < 0 & > 0 \\ \therefore (\sqrt{3}, \frac{2}{3\sqrt{3}}) & (-\sqrt{3}, \frac{-2}{3\sqrt{3}}) \\ \text{MAX} & \text{MIN} \end{array} \quad \checkmark$$

(e) Sketch the curve $y = \frac{x^2 - 1}{x^3}$, clearly labelling any stationary points, asymptotes, and intercepts with the axes.

4



$$\text{as } x \rightarrow \infty$$

$$y \rightarrow 0^+$$

$$\text{as } x \rightarrow -\infty$$

$$y \rightarrow 0^-$$

$$\text{as } x \rightarrow 0^-$$

$$y \rightarrow \infty$$

$$\text{as } x \rightarrow 0^+$$

$$y \rightarrow -\infty$$

QUESTION THIRTY-FOUR (3 marks)

Marks

Solve $2 \log_2 (x + 3) - \log_2 x = 4$.

3

$$\log_2 \left(\frac{(x+3)^2}{x} \right) = 4 \quad \checkmark$$

$$\frac{(x+3)^2}{x} = 2^4$$

$$(x+3)^2 = 16x$$

$$x^2 + 6x + 9 = 16x$$

$$x^2 - 10x + 9 = 0 \quad \checkmark$$

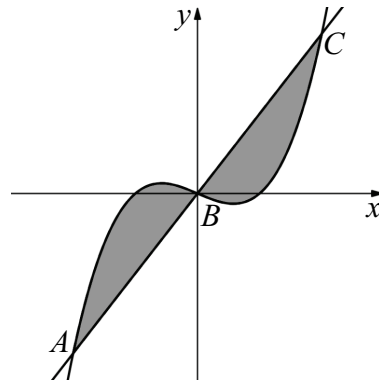
$$(x-9)(x-1) = 0$$

$$x = 9 \quad \text{or} \quad x = 1 \quad \checkmark$$

QUESTION THIRTY-FIVE (4 marks)

Marks

4



Given $f(x) = x^3 - x$ and $g(x) = 3x$, find the area enclosed by $y = f(x)$ and $y = g(x)$, as shown by the shaded regions in the diagram above.

intersect when

$$f(x) = g(x)$$

$$x^3 - x = 3x$$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x(x+2)(x-2) = 0$$

$$\therefore \text{EITHER } x=0 \quad x=-2 \quad x=2$$

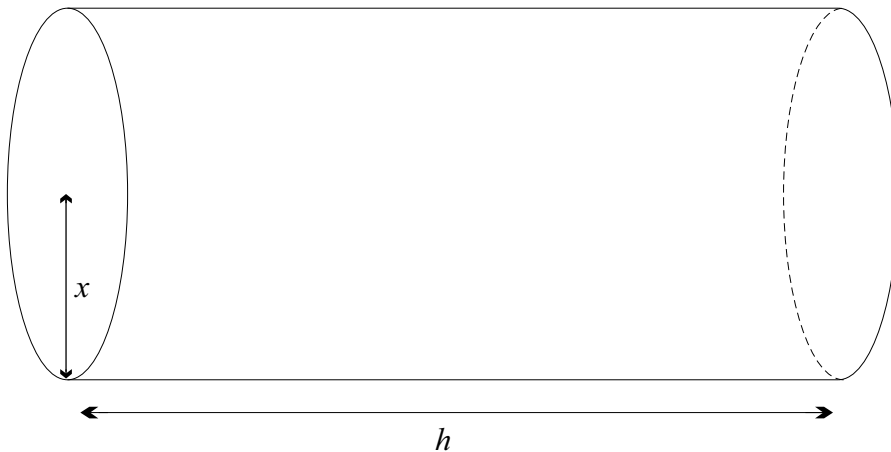
both functions odd $\therefore A = 2 \int_0^2 3x - (x^3 - x) dx$

$$= 2 \int_0^2 4x - x^3 dx$$

$$= 2 \left[2x^2 - \frac{x^4}{4} \right]_0^2$$

$$= 2(8 - 4) - 0$$

$$= \underline{\underline{8 \text{ units}^2}}$$

QUESTION THIRTY-SIX (6 marks)**Marks**

The diatium power cell is an integral component in all lightsabers. Each cell is cylindrical in shape, with a base radius x cm and height h cm, as shown in the diagram above.

- (a) Given that the volume of each cell has to be 60 cm^3 , show that the surface area, $A\text{ cm}^2$, of the cell is given by $A = 2\pi x^2 + \frac{120}{x}$. 2

$$V = \pi r^2 h$$

$$60 = \pi x^2 h$$

$$h = \frac{60}{\pi x^2}$$

✓

$$A = 2\pi r^2 + 2\pi r h$$

$$= 2\pi x^2 + 2\pi x \times \frac{60}{\pi x^2}$$

✓ show that

$$A = 2\pi x^2 + \frac{120}{x} \quad \text{As required.}$$

QUESTION THIRTY-SIX (Continued)

- (b) For safety reasons it is imperative that the surface area is as low as possible. Calculate the minimum surface area, correct to 2 decimal places, and justify your answer.

4

$$A = 2\pi x^2 + 120x^{-1}$$

$$\frac{dA}{dx} = 4\pi x - 120x^{-2}$$

$$\frac{d^2A}{dx^2} = 4\pi + 240x^{-3}$$

s.p. where $\frac{dA}{dx} = 0$

$$4\pi x = \frac{120}{x^2} \quad x \neq 0$$

$$x^3 = \frac{120}{4\pi}$$

$$x = \sqrt[3]{\frac{30}{\pi}}$$

$$\frac{d^2A}{dx^2} = 4\pi + \frac{240}{\frac{120}{4\pi}}$$

$$\doteq 37.699$$

$$> 0$$

$$\therefore \text{MIN}$$

$$A = 2\pi \left(\sqrt[3]{\frac{30}{\pi}} \right)^2 + \frac{120}{\sqrt[3]{\frac{30}{\pi}}}$$

$$\doteq 84.84 \text{ cm}^2$$

QUESTION THIRTY-SEVEN (5 marks)

Marks

Bo-Katan and Din are racing with their jetpacks along a 2km linear course. Bo-Katan accelerates at a constant 10m/s^2 up to a top speed of 35m/s , which she maintains until the finish. Din accelerates at a constant 8m/s^2 up to a top speed of 40m/s , which he maintains until the finish.

- (a) Due to a technical malfunction, Din starts n seconds after Bo-Katan. Given that they both start from rest, and Din wins the race, what is the largest possible value of n ? Give your answer correct to 2 decimal places.

3

Bo-Katan	$\frac{35}{10} = 3.5$	Din	$\frac{40}{8} = 5$
$a = 10$	$0 \leq t \leq 3.5$	$a = 8$	$0 \leq t \leq 5$
$v = 10t + c_1$	$0 \leq t \leq 3.5$	$v = 8t + c_4$	$0 \leq t \leq 5$
start from rest $c_1 = 0$		start from rest $c_4 = 0$	
$v = 10t$	$0 \leq t \leq 3.5$	$v = 8t$	$0 \leq t \leq 5$
$x = 5t^2 + c_2$		$x = 4t^2 + c_5$	
start from origin $c_2 = 0$		start from origin $c_5 = 0$	
$x = 5t^2$	$0 \leq t \leq 3.5$	$x = 4t^2$	$0 \leq t \leq 5$
$v = 35$	$3.5 < t$	$v = 40$	$5 < t$
$x = 35(t-3.5) + c_3$	$3.5 < t$	$x = 40(t-5) + c_6$	$5 < t$
when $t = 3.5$, $x = 5(3.5)^2 = 61.25$		when $t = 5$, $x = 4(5)^2 = 100$	
$x_{\text{BK}} = 35(t-3.5) + 61.25$	$3.5 < t$	$x_{\text{D}} = 40(t-5) + 100$	$5 < t$
$= 35t - 122.5 + 61.25$		$= 40t - 200 + 100$	
$= 35t - 61.25$	✓	$= 40t - 100$	✓
$2000 = 35t - 61.25$		$2000 = 40t - 100$	
$t = \frac{2000 + 61.25}{35}$		$t = \frac{2000 + 100}{40}$	
$\div 58.89...$		$= 52.5$	
$n = 58.89 - 52.5$			
$= 6.39\text{s}$	✓		

Alternate geometric approach below...

QUESTION THIRTY-SEVEN (Continued)

- (b) Suppose Din and Bo-Katan start at the same point at the same time, that is $n = 0$ seconds. 2

At what displacement, x , from the start of the race will they meet again?

In same place when $x_{Bk} = x_D$

$$38t - 61.25 = 40t - 100$$

$$38.75 = 5t$$

$$t = 7.75_s \quad \checkmark$$

$$x = 40(7.75 - 5) + 100$$

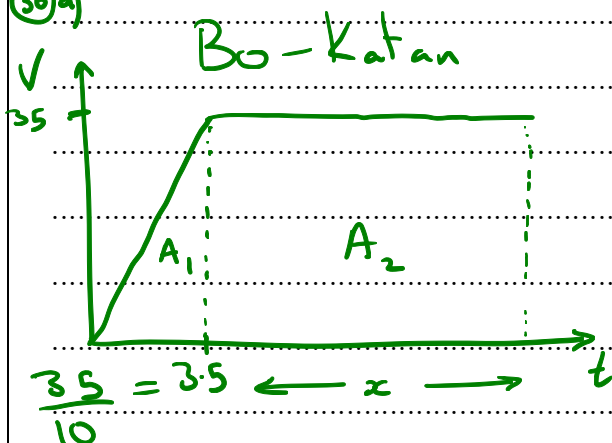
$$= 210m \quad \checkmark$$

————— END OF PAPER —————

Extra writing space (Use this space only for questions in Part D)

If you use this space, clearly indicate which question you are answering.

(36a)



$$A_1 = \frac{3.5 \times 35}{2}$$

$$= 61.25$$

$$A_2 = 2000 - 61.25$$

$$= 1938.75$$

$$x = \frac{1938.75}{35}$$

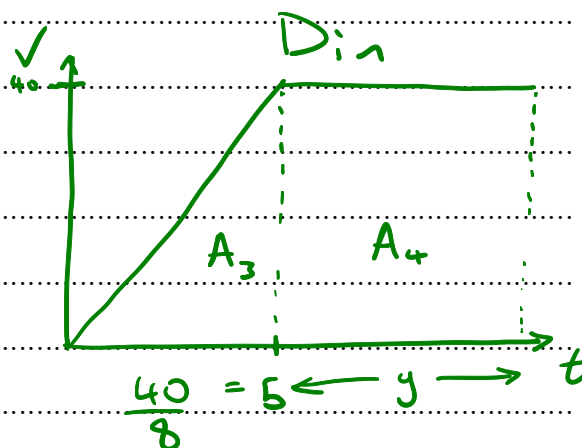
$$\div 55.39$$

$$\therefore \text{Bo-Katan takes } 55.39 + 3.5$$

$$= 58.89s$$

$$\therefore t = 58.89 - 52.5$$

$$= 6.39s$$



$$A_3 = \frac{5 \times 40}{2}$$

$$= 100$$

$$A_4 = 2000 - 100$$

$$= 1900$$

$$y = \frac{1900}{40}$$

$$y = 47.5$$

$$\therefore \text{Din takes } 47.5 + 5$$

$$= 52.5s$$